

Theme Description generation — EDA

Question: is the generated Jira Theme Description **grounded** (no invention) and **complete** (covers the ticket) — and can prompt work improve it?

Why this eval is reference-free

Description is **generative**, not classification — there is no single correct answer. The obvious reference is the BA's Jira description, but **each GT description is free-form with its own format**, so matching the generated text against it statement-by-statement would penalise **style, not substance** (the BA chose a different structure / emphasis, not different facts).

So we do **not** score against the GT. We judge each generated description **only against the source it was generated from** — the raw idea card. This is the ragas methodology (claim decomposition + grounding) implemented natively (no ragas/langchain dependency; works with the IDP gateway, and surfaces the offending claims). The GT file is used only to know **which value streams** to generate for.

What is generated, and what we judge

Theme generation, per ticket, uses the **raw idea card only** (no summary, no generation signals - consistent with the locked VS/stage decision):

- a shared **body** (one call) — product availability + initiative + capabilities, VS-agnostic;
- a per-VS **framing** paragraph (one batched call).

The final per-value-stream description = **framing + shared body, assembled**. We judge that **combined, assembled description per value stream** (not the body or a framing in isolation - the assembled text is the artifact the architect sees).

The three metrics (all vs the raw source)

metric	measures	direction
faithfulness	every claim is supported by the idea card (= "correctness" when reference-free: a claim is right iff the source backs it)	↑
hallucination	1 - faithfulness — the unsupported (invented) claims, listed	↓
coverage	the idea card's key facts reflected in the description (did it omit important content?)	↑

Faithfulness/hallucination = *don't make things up* (fact precision vs source); coverage = *don't leave things out* (fact recall vs source). A good description is **both**.

Where is "correctness"? There is no separate correctness metric, by design. Correctness-vs-GT was dropped because each GT description is free-form (matching it would penalise style, not substance). In a reference-free

eval, "is this claim correct?" becomes "is it supported by the source?" — so **faithfulness IS the correctness check**, and coverage is the completeness check. We keep faithfulness + hallucination + coverage rather than a redundant "correctness" label.

Finding — a grounding prompt halved hallucination

The first run was good but leaked invention:

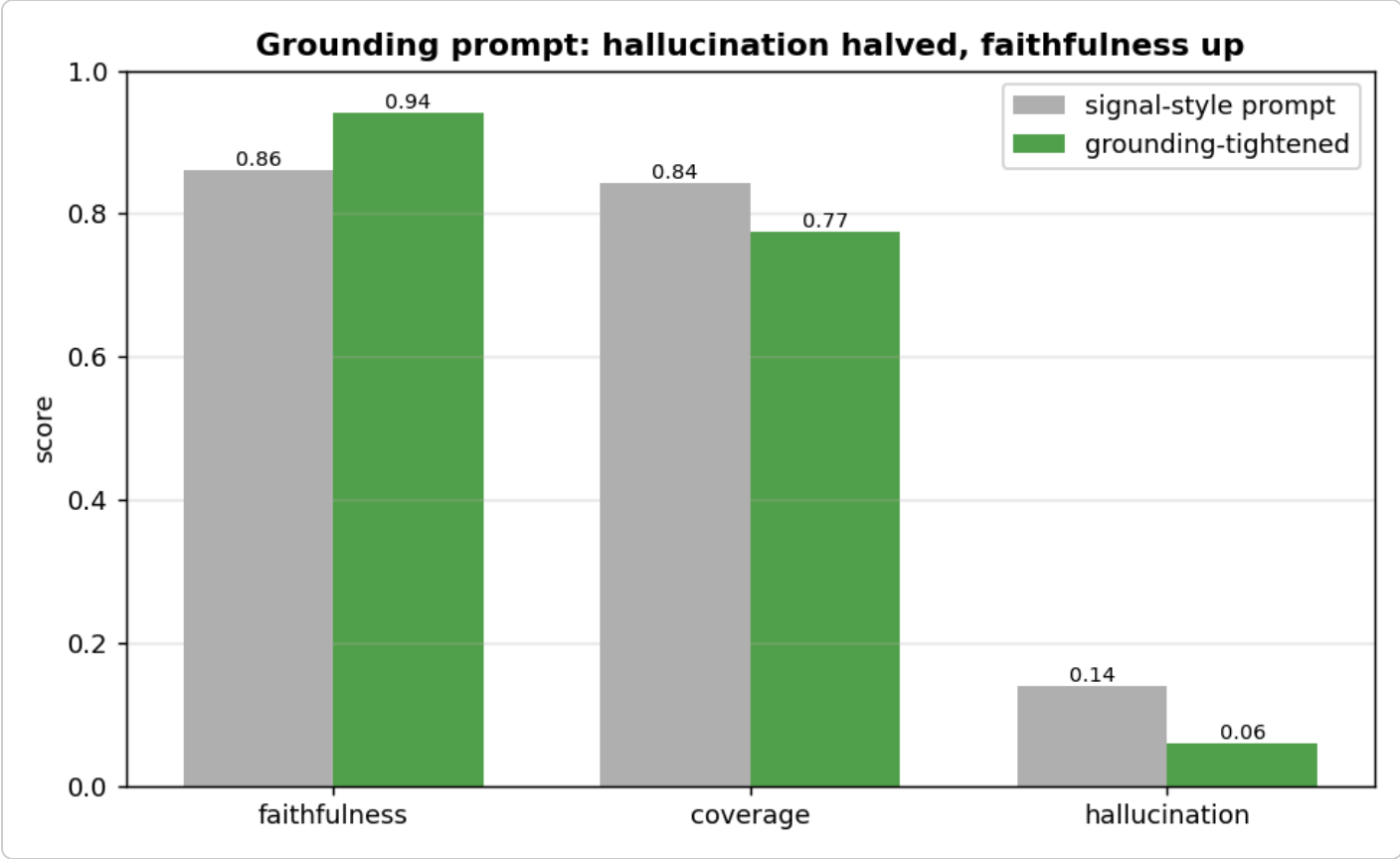
metric	baseline
faithfulness	0.860
hallucination	0.140
coverage	0.843
avg claims/desc	21.8
avg unsupported/desc	2.75

The last two are the **raw claim counts** behind faithfulness/hallucination: the faithfulness judge decomposes each description into atomic factual claims, so `avg claims/desc = 21.8` means each description asserts ~22 self-contained facts, and `avg unsupported/desc = 2.75` means ~3 of those 22 aren't supported by the source. Faithfulness/hallucination are just these as ratios (`supported/total` and `unsupported/total`), but the counts are more concrete — "*~3 of ~22 claims invented*" is easier to picture than *0.14*, and the count is what the prompt fix moves directly.

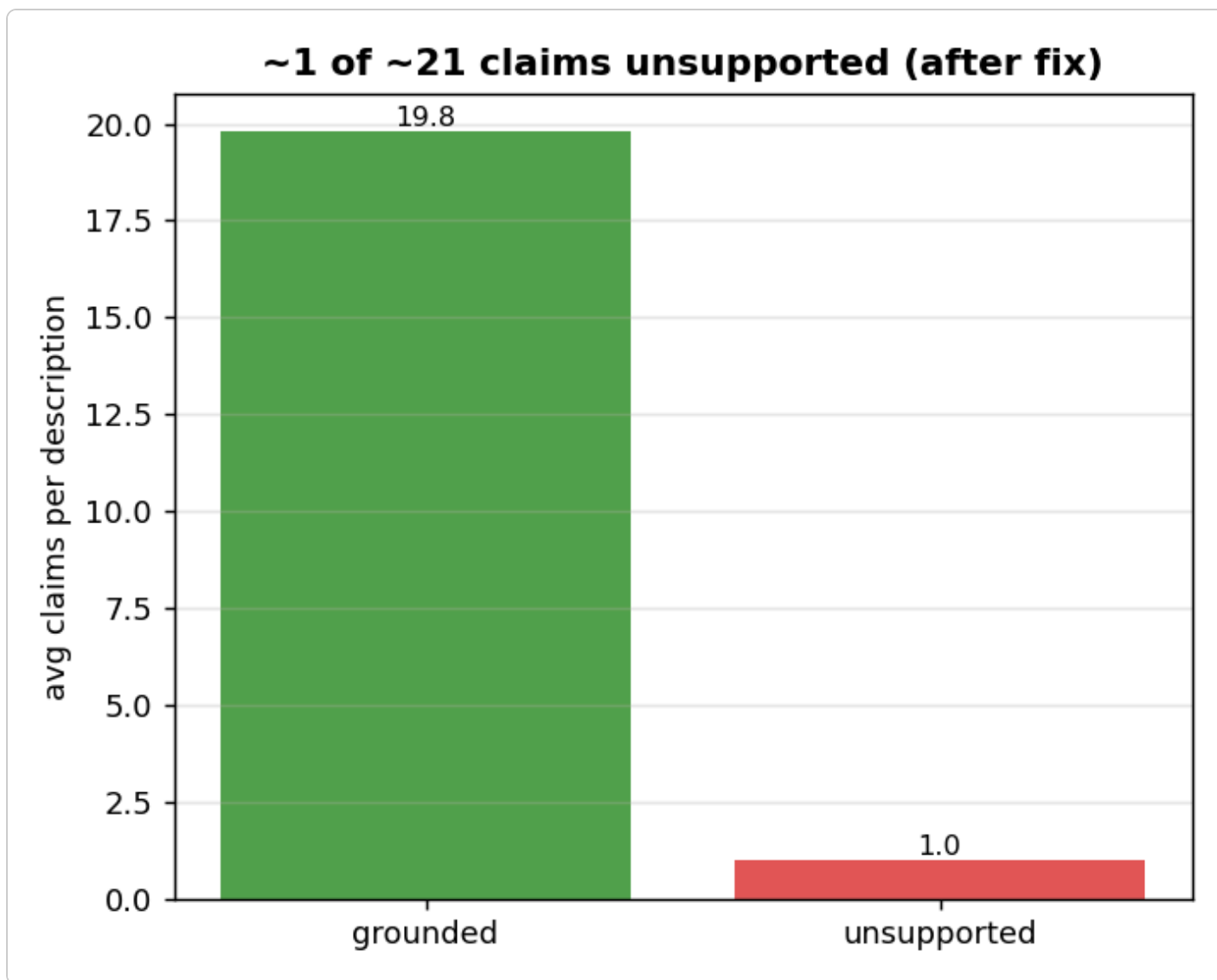
Diagnosis: the body prompt was **signal-centric** — it told the model to "copy the value from the matching signal" for Product Availability / funding / networks. But theme generation now feeds the **raw idea card with no signals**, so the model had nothing to copy and **inferred those structured facts from narrative = invention**. ~3 of every ~22 claims weren't grounded.

Fix (both body + framing prompts):

- lead with a hard **GROUNDING rule**: every statement must be supported by a specific phrase in the idea card; if you can't point to it, omit it; *a shorter grounded description beats a padded one*;
- Product Availability fields: include a line **only when the card explicitly states it** (never infer dates / plans / funding / networks);
- capability/integration bullets must name something the card describes — no "standard" capabilities;
- framing: don't invent stakeholders / outcomes / scope to fill the paragraph.



metric	before	after	Δ
faithfulness	0.860	0.940	+0.08
hallucination	0.140	0.060	-57%
coverage	0.843	0.774	-0.07
avg unsupported/desc	2.75	0.98	-64%



How to read it: unsupported claims fell from ~2.75 per description to ~1 — invention went from *1 in 7 claims* to *1 in 17*. Faithfulness 0.94 is a strong profile for a business artifact a BA signs off on.

The trade — and why it's the right one

Tightening grounding cost **~7 points of coverage** (0.84 → 0.77): "when in doubt, leave it out" omits a bit more peripheral detail. For a **trust-critical** artifact this is a good trade: **inventing a fact (a funding model, a go-live date) is far worse than omitting a minor one** — a BA can add a missed point, but can't trust a document that fabricates. No-invention is the domain's #1 rule, so we optimise faithfulness over completeness here.

Verdict — locked

Theme Description: raw idea card only, grounding-tightened prompts → faithfulness 0.94 / hallucination 0.06 / coverage 0.77.

- Reference-free, judged against the source (the GT format is too varied to score against).
- The combined framing + body description is judged per value stream.
- Prompt grounding was the lever: hallucination halved, faithfulness up, at a small coverage cost we accept because invention is the worse failure for a BA-facing artifact.

These are conservative floor numbers. Spot-checks showed the automated judge occasionally marks source-grounded claims as unsupported and counts correctly-omitted project financials as missed coverage, so true faithfulness is higher and true hallucination lower than shown — the real quality is better than the metrics suggest.

No further changes.